

Travaux pratiques - Tests automatisés utilisant PYATS et Genie

Etape 3

a) Voici l'environnement virtuel

```
devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ ls -l
total 20
drwxrwxr-x 2 devasc devasc 4096 May 28 19:12 bin
drwxrwxr-x 2 devasc devasc 4096 May 28 19:12 include
drwxrwxr-x 3 devasc devasc 4096 May 28 19:12 lib
lrwxrwxrwx 1 devasc devasc    3 May 28 19:12 lib64 -> lib
-rw-rw-r-- 1 devasc devasc  69 May 28 19:12 pyenv.cfg
drwxrwxr-x 3 devasc devasc 4096 May 28 19:12 share
devasc@labvm:~/labs/devnet-src/pyats/csr1kv$
```

b)

```
devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ cat pyenv.cfg
home = /usr/bin
include-system-site-packages = false
version = 3.8.2
```

je vois bien que le fichier pointe vers l'emplacement de l'installation Python

c)

```
devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ ls -l /usr/bin/python*
/usr/bin/python3
/usr/bin/python3.8
/usr/bin/python3.8-config
/usr/bin/python3-config
/usr/bin/python-argcomplete-check-easy-install-script3
/usr/bin/python-argcomplete-tcsh3
devasc@labvm:~/labs/devnet-src/pyats/csr1kv$
```

e)

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$
```

Partie 3 : Utiliser la bibliothèque de tests PyATS

Étape 2: Verifying pyATS.

PyATS est bien installé:

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ pyats --help
Usage:
  pyats <command> [options]

Commands:
  clean          runs the provided clean file
  create         create scripts and libraries from template
  develop        Puts desired pyATS packages into development
  diff           Command to diff two snapshots saved to file
  dnac           Command to learn DNAC features and save to
```

Étape 3: Cloner et examiner les exemples de scripts PyATS de GitHub.

a)

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ git clone https://github.com/CiscoTestAutomation/examples
Cloning into 'examples'...
remote: Enumerating objects: 1506, done.
remote: Counting objects: 100% (354/354), done.
remote: Compressing objects: 100% (184/184), done.
remote: Total 1506 (delta 169), reused 252 (delta 140), pack-reused 1152 (from 1)
Receiving objects: 100% (1506/1506), 1.21 MiB | 4.99 MiB/s, done.
Resolving deltas: 100% (740/740), done.
```

b)

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ ls -l
total 24
drwxrwxr-x  3 devasc devasc 4096 May 28 19:31 bin
drwxrwxr-x 25 devasc devasc 4096 May 28 19:48 examples
drwxrwxr-x  2 devasc devasc 4096 May 28 19:12 include
drwxrwxr-x  3 devasc devasc 4096 May 28 19:12 lib
lrwxrwxrwx  1 devasc devasc   3 May 28 19:12 lib64 -> lib
-rw-rw-r--  1 devasc devasc  69 May 28 19:12 pyenv.cfg
drwxrwxr-x  3 devasc devasc 4096 May 28 19:12 share
```

c)

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ ls -l examples/
total 100
drwxrwxr-x  3 devasc devasc 4096 May 28 19:48 abstraction_example
drwxrwxr-x  3 devasc devasc 4096 May 28 19:48 basic
drwxrwxr-x  3 devasc devasc 4096 May 28 19:48 blitz
drwxrwxr-x 31 devasc devasc 4096 May 28 19:48 clean
drwxrwxr-x  7 devasc devasc 4096 May 28 19:48 comprehensive
drwxrwxr-x  4 devasc devasc 4096 May 28 19:48 connection
drwxrwxr-x  5 devasc devasc 4096 May 28 19:48 datafiles
drwxrwxr-x  3 devasc devasc 4096 May 28 19:48 feature
drwxrwxr-x  2 devasc devasc 4096 May 28 19:48 GenieHarnessHelloWorld
drwxrwxr-x  2 devasc devasc 4096 May 28 19:48 groups
drwxrwxr-x  4 devasc devasc 4096 May 28 19:48 health
drwxrwxr-x 15 devasc devasc 4096 May 28 19:48 libraries
-rw-rw-r--  1 devasc devasc 11357 May 28 19:48 LICENSE
drwxrwxr-x  2 devasc devasc 4096 May 28 19:48 loop
```

d)

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ ls -l examples/basic
total 16
-rw-rw-r--  1 devasc devasc  534 May 28 19:48 basic_example_job.py
-rwxrwxr-x  1 devasc devasc 4510 May 28 19:48 basic_example_script.py
drwxrwxr-x  2 devasc devasc 4096 May 28 19:48 results
```

Étape 4: Examinez les fichiers de script de base.

a)

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ cat examples/basic/basic_e
xample_script.py | more
#!/usr/bin/env python
#####
# basic_example.py : A very simple test script example which include:
#   common_setup
#   Tescases
#   common_cleanup
# The purpose of this sample test script is to show the "hello world"
# of aetest.
#####
# To get a logger for the script
import logging

# Needed for aetest script
from pyats import aetest
```

b)

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ cat examples/basic/basic_example_job.py
"""
<PYATS_JOBFILE>
"""
# To run the job:
# pyats run job basic_example_job.py
# Description: This example shows the basic functionality of pyats
#              with few passing tests

import os
from pyats.easypy import run

# All run() must be inside a main function
def main():
    # Find the location of the script in relation to the job file
    test_path = os.path.dirname(os.path.abspath(__file__))
    testscript = os.path.join(test_path, 'basic example script.py')
```

Partie 4 : Utiliser Genie pour analyser la sortie de la commande IOS

a)

```
(csr1kv) devasc@labvm:~/labs/devnet-src/pyats/csr1kv$ genie --help
Usage:
  genie <command> [options]

Commands:
  create          Create Testbed, parser, triggers, ...
  develop         Puts desired pyATS packages into development mode
  diff            Command to diff two snapshots saved to file or directory
  dnac            Command to learn DNAC features and save to file (Prototype)
  learn           Command to learn device features and save to file
  parse           Command to parse show commands
  run            Run Genie triggers & verifications in pyATS runtime environment
  shell           enter Python shell, loading a pyATS testbed file and/or pickled data
  undevelop       Removes desired pyATS packages from development mode
```